Public Perceptions of Algorithmic vs. Human-Drawn Congressional Districts: A Survey Experiment

Giovanni Della-Libera

Working Paper

May 5, 2026

Abstract

Gerrymandering is widely condemned by the American public, yet political reform remains slow. Algorithmic redistricting—computer-drawn district maps that use no partisan data—offers a structural alternative, but public acceptance depends on perceived legitimacy. We conducted a pre-registered survey experiment ($N = 2{,}400$) using a 2×3 between-subjects design crossing map type (algorithmic, commission-drawn, or enacted) with information condition (none, process description, or both). Respondents rated algorithmic maps 0.41 standard deviations fairer than enacted maps ($p < 0.001$). Providing a plain-language description of the algorithm raised perceived fairness by an additional 0.22 SD ($p < 0.001$). Partisan moderation was smaller than anticipated: Democrats rated algorithmic maps 0.18 SD higher than Republicans, a statistically significant but substantively modest gap ($p = 0.04$). Critically, no significant difference emerged between algorithmic and commission-drawn maps. We find no evidence of algorithm aversion in this domain; to the contrary, transparency about algorithmic process functions as a legitimacy-enhancing treatment. These results suggest that public support for algorithmic redistricting is achievable across partisan lines when process descriptions accompany map presentation.

Keywords: redistricting, gerrymandering, survey experiment, algorithm aversion, procedural fairness, public opinion

1 Introduction

Americans across partisan lines have long condemned gerrymandering. Surveys consistently find that majorities of Democrats, Republicans, and independents view partisan manipulation of district lines as unfair (Pew Research Center, 2017; Gallup Organization, 2019). Yet the gap between public disapproval and institutional reform is wide and persistent. As of the 2020 redistricting cycle, most states continued to assign line-drawing authority to legislatures with undisguised partisan interests in the outcome. The Supreme Court's decision in *Rucho v. Common Cause* (2019) removed the federal judiciary as a check on partisan gerrymandering, leaving structural reform to state legislatures, citizen initiatives, and state courts—all of which move slowly.

One structural alternative has received increasing scholarly and journalistic attention: algorithmic redistricting. Under this approach, a computer program draws district boundaries by optimizing objective criteria—population equality, geographic contiguity, and compactness—without accessing any data about voters' party affiliation, race, or voting history. The algorithm is open-source, reproducible, and structurally incapable of implementing partisan strategy because it cannot see partisan data (Della-Libera, 2026). Where human mapmakers can exploit every census tract's partisan lean to engineer favorable outcomes, algorithmic mapmakers cannot. The impossibility of manipulation is not a normative aspiration but a mathematical property of the process.

Despite growing technical validation of algorithmic approaches, a critical question remains unanswered: *do ordinary citizens find algorithmic maps legitimate?* Algorithmic redistricting carries a different kind of risk than legislative or even commission-based redistricting—not the risk of partisan manipulation, but the risk of algorithm aversion, a well-documented tendency for people to discount or distrust algorithmic judgments even when those judgments demonstrably outperform human alternatives (Dietvorst et al., 2015). If the public perceives algorithmic maps as cold, opaque, or undemocratic—as decisions "made by a machine" rather than by accountable humans—then technical superiority may not translate into political feasibility.

This paper addresses the legitimacy gap directly with a large-scale survey experiment. We recruited $N = 2{,}400$ respondents from the MTurk and Lucid platforms, pre-registered our hypotheses and analysis plan on the Open Science Framework, and randomly assigned participants to view one of three types of congressional district maps—algorithmic, commission-drawn, or enacted—in one of three information conditions: no description of the process, a plain-language description of how the map was produced, or both the description and an explicit comparison to the alternative. We then measured perceived fairness, trust in the redistricting process, partisan bias perceptions, and willingness to accept the map's outcomes.

Our results challenge several priors simultaneously. First, we find robust positive public perceptions of algorithmic maps: respondents rated them 0.41 standard deviations fairer than enacted maps, an effect that is large by the standards of survey experiments on redistricting (Bonneau and Cann, 2021; Elmendorf and Quinn, 2019): Second, we find that process description substantially amplifies this advantage, adding 0.22 SD to perceived fairness ratings. This transparency-as-legitimacy finding echoes procedural fairness theory (Thibaut and Walker, 1975; Tyler, 2006): knowing *how* a decision was made shapes perceptions of *whether* it is fair, independent of the outcome. Third, partisan heterogeneity is smaller than expected. Democrats rated algorithmic maps higher than Republicans by 0.18 SD—a statistically significant but substantively modest difference—suggesting that the appeal of algorithmic redistricting crosses party lines when process information is provided. Fourth, and perhaps most policy-relevant, we find no significant difference between algorithmic and commission-drawn maps in perceived fairness. If the public treats these two reform alternatives as equivalent in legitimacy, then algorithmic redistricting passes the democratic acceptance threshold already established by the independent commission model.

The paper proceeds as follows. Section 2 reviews the literatures on redistricting opinion, procedural fairness, and algorithm aversion that motivate our hypotheses. Section 3 describes the 2×3 experimental design, sampling strategy, and pre-registration. Section 4 presents the outcome measures and their validation. Section 5 reports the main experimental findings, including heterogeneous effects by partisanship, education, and political knowledge. Section 6 discusses implications for reform politics and compares our findings to prior work. Section 7 concludes with policy recommendations.

1.1 The Legitimacy Problem in Redistricting Reform

Understanding why reform has been so difficult requires distinguishing between two types of legitimacy that redistricting maps must satisfy. *Outcome legitimacy* concerns whether the resulting districts are judged as fair—whether seat distributions roughly correspond to vote shares, whether district shapes are recognizable as communities, whether minority representation is preserved. *Process legitimacy* concerns whether the procedure that produced the map was itself fair—transparent, neutral, free from self-interested manipulation, and open to public input (Tyler, 2006).

American redistricting reform efforts have focused heavily on process legitimacy: the independent commission model attempts to move mapmaking authority away from self-interested legislatures toward bodies composed of citizens without partisan stakes in the outcome. Arizona, California, Michigan, and Colorado all adopted commission-based reforms precisely on procedural grounds. The empirical evidence on whether commissions actually produce better outcomes is mixed (Imig and Imig, 2022; Kousser, 2013; Stephanopoulos and Warshaw, 2015): commissions reduce the most egregious partisan bias but do not eliminate it, and they introduce their own procedural tensions around commissioner selection and public input weighting.

Algorithmic redistricting addresses process legitimacy in a structurally different way. Rather than substituting one set of human decision-makers for another, it substitutes a mathematical procedure for human line-drawing. The algorithm cannot be captured by partisan interests because it has no access to partisan data. Its outputs are reproducible—the same inputs always produce the same map—which enables independent verification in a way that commission deliberations cannot. Its decision process is fully documented in open-source code, not in meeting minutes or commissioner testimony.

Whether these procedural properties translate into public perceived legitimacy is the empirical question this paper answers.

2 Related Work

Our study draws on three bodies of literature: public opinion research on redistricting and gerrymandering, procedural fairness theory, and the psychology of algorithm aversion.

2.1 Public Opinion on Redistricting

Public knowledge of and opinion about redistricting has been studied systematically for roughly two decades (Henderson, Haworth and Kim, 2018; Bandyopadhyay and Oak, 2013; Clark and Leiter, 2021). Several consistent findings emerge. First, most Americans have low baseline knowledge of how congressional districts are drawn: surveys typically find fewer than one in three respondents can correctly identify who draws congressional district lines in their state (Gimpel and Tam Cho, 2020; Henderson, Haworth and Kim, 2018). Second, when provided with descriptions of gerrymandering—including images of distorted districts—majorities across partisan groups rate the practice as unfair (Elmendorf and Quinn, 2019; Nalder, 2019). Third, public preferences for redistricting process strongly favor independent commissions over legislative control, with commission support in the 60–70% range across multiple polls (Pew Research Center, 2017; Gallup Organization, 2019).

Less studied is public opinion about *algorithmic* redistricting specifically. Nalder (2019) found that subjects shown maps described as "drawn by computer" rated them as more fair than maps described as "drawn by the legislature," but the study used a small convenience sample and did not vary information about the algorithmic process. Henderson, Haworth and Kim (2018) asked respondents whether they would support computer-drawn districts and found plurality support (approximately 48%), but did not experimentally test whether exposure to actual maps or process descriptions affects these preferences. Our study fills this gap with a pre-registered experiment at scale.

The partisan asymmetry in redistricting perceptions is well-documented. Partisans evaluate maps more favorably when those maps benefit their party (Clark and Leiter, 2021; Bonneau and Cann, 2021). This motivated hypothesis testing party as a moderator—if algorithmic maps are perceived as taking away partisan advantage, the out-party (whichever party benefits less from the current enacted map) should show larger positive responses to the algorithmic alternative. Our design tests this by focusing on enacted maps in states with known partisan gerrymanders favoring each party.

Importantly, research on redistricting reform support suggests that institutional features other than partisan advantage also shape opinion. Imig and Imig (2022) showed that transparency and public accountability in commission processes increase support for commission-drawn maps. Stephanopoulos and Warshaw (2015) found that detailed process descriptions—explaining the criteria commissions use and how they apply them—raised fairness ratings substantially. These findings set the stage for our transparency treatment.

2.2 Procedural Fairness Theory

Procedural fairness theory, developed by Thibaut and Walker (1975) and elaborated by Tyler (2006, 1988), holds that people's evaluations of outcomes are powerfully shaped by their assessments of the *process* that produced those outcomes. Even unfavorable outcomes may be accepted as legitimate if the procedure was perceived as fair—neutral, transparent, voice-providing, and consistent. Conversely, favorable outcomes may be rejected if the procedure was seen as biased or opaque.

Four procedural characteristics consistently emerge as central to legitimacy assessments (Leventhal, 1980; Lind and Tyler, 1988): *consistency* (the same rules applied equally to all); *bias suppression* (decision-makers lack self-interest in the outcome); *accuracy* (decisions are based on correct information); and *correctability* (errors can be challenged and revised). Algorithmic redistricting has structural advantages on the first two dimensions—algorithms are definitionally consistent, and properly designed algorithms can be made free of partisan self-interest. These properties should translate into procedural legitimacy perceptions when subjects understand the process.

Applied to redistricting, Gerber and Lewis (2018) showed that respondents in court-ordered redistricting states rated the resulting maps as more procedurally fair, despite often rating them as less favorable to their party's interests. This finding supports the prediction that process neutrality, when perceived, generates legitimacy independent of outcome. Our transparency treatment tests whether communicating process neutrality in plain language achieves the same effect in a controlled experimental setting.

A limitation in applying procedural fairness theory to algorithmic contexts is that algorithms may be perceived as *non-human* in ways that undercut the typical legitimacy pathway. Standard procedural fairness research involves human decision-makers; respondents may extend different standards to algorithmic procedures (Svensson, 2022). The voice component of procedural fairness—the belief that one had an opportunity to be heard—is structurally different when the decision-maker is code rather than a commissioner. We address this concern in the discussion.

2.3 Algorithm Aversion

A substantial literature documents *algorithm aversion*: the tendency for people to distrust algorithmic judgments and prefer human judgment even when algorithms demonstrably outperform humans (Dietvorst et al., 2015; Dietvorst, Logg and Massey, 2018; Logg, Minson and Moore, 2019). Dietvorst et al. (2015) showed that subjects who watched an algorithm make errors subsequently preferred human forecasters over the algorithm, even when the algorithm's predictions were objectively more accurate. The effect appears to stem from a belief that algorithms are rigid and cannot adapt to exceptional cases—a concern that human experts, by contrast, can exercise contextual judgment.

Several factors moderate algorithm aversion. Logg, Minson and Moore (2019) showed that algorithm aversion reverses to *algorithm appreciation* when subjects are experts in the relevant domain—that is, experts trust algorithms more than novices do. Dietvorst, Logg and Massey (2018) found that allowing subjects to adjust algorithmic outputs substantially reduced aversion, presumably because it preserved the sense of human agency in the final outcome. Castelo, Bos and Lehmann (2019) found that algorithm aversion is stronger for tasks perceived as requiring uniquely human capacities (e.g., social judgment) than for tasks perceived as objective computation. Redistricting, as a geographic optimization problem with mathematically specifiable criteria, may be perceived as closer to the latter category.

The existing algorithm aversion literature is largely silent on political redistricting. The one domain closest to our context is algorithmic hiring and personnel decisions, where Lee (2018) found that perceived fairness of algorithmic decisions was sharply moderated by transparency: subjects who received explanations of how hiring algorithms worked rated them as substantially more fair than subjects who received only the output. This transparency-fairness link motivates our information condition manipulation.

Our study makes three contributions to the algorithm aversion literature. First, we test whether aversion appears in a politically salient domain where the relevant comparison is not algorithm-vs.-human expert but algorithm-vs.-politically-motivated legislature. Second, we test whether plain-language process descriptions reduce or eliminate any aversion that appears. Third, we test partisan moderation of aversion—whether political stakes create systematic differences in algorithm evaluations between partisan groups.

2.4 Hypotheses

Drawing on these three literatures, we pre-registered the following hypotheses:

H1 (Main effect): Respondents will rate algorithmic maps as fairer than enacted maps.

H2 (Transparency): Respondents will rate algorithmic maps no less fair than commission-drawn maps (null hypothesis of equivalence, tested via one-sided t-test). A plain-language description of the algorithmic process will increase perceived fairness for algorithmic and commission maps relative to the enacted baseline.

H3 (Exploratory): We explore whether partisan identification moderates fairness perceptions. Given per-subgroup sample sizes of $n \approx 130$–150, this analysis is underpowered to detect effects smaller than 0.36 SD at 80% power and should be treated as descriptive.

H4 (Algorithm vs. commission): Algorithmic and commission-drawn maps will not differ significantly in perceived fairness.

H5 (Process trust): Trust in the redistricting process will be higher for algorithmic than enacted maps, with transparency amplifying this effect.

3 Research Design

3.1 Experimental Design

We implemented a 2×3 between-subjects design crossing *map type* (algorithmic, commission-drawn, or enacted) with *information condition* (none, process description, or full description plus comparison). The six cells are shown in Table 1. Each respondent was randomly assigned to exactly one cell and saw two congressional district maps from two states—one where the enacted map favors Democrats (North Carolina) and one where it favors

Republicans (Maryland)—drawn in the assigned map-type condition. Presenting maps from both partisan directions allows us to assess whether fairness ratings depend on whether the enacted map disadvantages or advantages the respondent's party.

Table 1: 2×3 Experimental Design

	Information Condition		
Map Type	None	Process Description	Description + Comparison
Algorithmic	Cell A ($n = 400$)	Cell B ($n = 400$)	Cell C ($n = 400$)
Commission-drawn	Cell D ($n = 400$)	Cell E ($n = 400$)	Cell F ($n = 400$)
Enacted	(reference) ($n = 400$)	–	–

Note: The enacted/none cell serves as reference. Enacted maps with information conditions were excluded to avoid directing attention to the maps' provenance in ways that confound treatment effects. $N = 2{,}400$ total.

Maps were presented without labels indicating their type. Respondents in the no-information condition saw only the map image and basic factual information (state name and number of congressional districts). Respondents in the process-description condition saw the same map plus a 90-word plain-language description of how the map was produced. Respondents in the full-description-plus-comparison condition saw the process description and a second-panel image of the alternative map type (commission or algorithmic) with a brief explanation of how the two processes differ.

3.2 Map Stimuli

Algorithmic maps were generated using the recursive bisection algorithm described in Della-Libera (2026) applied to 2020 Census tract data for North Carolina (14 districts) and Maryland (8 districts). Commission-drawn maps used the final plans adopted by the North Carolina Bipartisan Redistricting Committee (2022) and the Maryland Citizens Redistricting Commission (2022). Enacted maps were the legislatively approved plans in force for the 2022 elections in each state.

All maps used a consistent visual design: district boundaries in dark gray on a white

Treatment Text Administered to Respondents

Process-description condition (algorithmic map):

> This congressional district map was drawn by a computer algorithm that minimizes the total length of district borders while ensuring equal population. The algorithm uses no information about voters' political preferences or racial identity. It starts from census tract boundaries and divides them into districts by finding the geographic partition that produces the most compact shapes possible, given that each district must contain the same number of people.

Process-description condition (commission-drawn map):

> This congressional district map was drawn by a bipartisan citizens' commission. Commission members were selected to represent diverse geographic and demographic backgrounds. The commission held public hearings in each region of the state and applied criteria including equal population, contiguity, compactness, and preservation of communities of interest. No sitting elected official or candidate for office was permitted to serve on the commission.

Figure 1: Exact process-description treatment text administered to respondents. The no-information condition showed only the map image and basic factual information (state name, number of districts). The full-description-plus-comparison condition appended a second-panel image of the alternative map type with a brief explanation of how the two processes differ.

background, districts filled with one of seven distinguishable colors (not red-blue), and no partisan or demographic overlays. A professional cartographic designer produced the final map images at 1400×900 pixels. The design was pre-registered and finalized before any respondent viewed stimuli.

3.3 Sampling and Recruitment

We recruited respondents through two platforms: Amazon Mechanical Turk (MTurk; $n =$ 1,200) and Lucid Theorem ($n =$ 1,200). MTurk provides a sample that is more educated and politically engaged than the general population but is commonly used in political science experiments (Berinsky, Huber and Lenz, 2012). Lucid uses quota sampling to match the

American adult population on age, gender, education, and region and has been shown to produce results generalizable to population-representative surveys (Coppock and McClellan, 2019). Recruiting from both platforms allows us to test whether results replicate across samples with different selection properties.

Quota targets for the Lucid sample were: 50% female; ages 18–29 (22%), 30–44 (25%), 45–64 (34%), 65+ (19%); education below high school diploma (10%), high school diploma (28%), some college (30%), four-year degree or more (32%); region Northeast (18%), Midwest (21%), South (38%), West (23%). We did not quota on party identification to allow natural variation.

A priori power analysis assumed a minimum detectable effect of $d = 0.20$ standard deviations on the primary outcome, $\alpha = 0.05$, and power $= 0.80$. This required $n = 394$ per cell, which we rounded to 400 per cell. The design was pre-registered on OSF at `https://osf.io/[redacted]` before data collection.

3.4 Pre-Registration

All hypotheses, analysis decisions, and stopping rules were pre-registered before survey launch. Pre-registered deviations from the analysis plan are noted inline where they occur. The pre-registration specified: (1) the primary outcome variable and its operationalization; (2) the regression specification, including control variables; (3) subgroup analyses by party identification, education, and political knowledge; (4) exclusion criteria for failed attention checks; and (5) the threshold for declaring the transparency treatment effective ($p < 0.05$, one-tailed, for the interaction of map type and information condition).

We also pre-registered that, if MTurk and Lucid samples produced substantively divergent results (standardized effect-size difference > 0.30), we would report them separately rather than pooling. They did not.

3.5 Procedure

The survey proceeded in four stages. In Stage 1 (approximately 3 minutes), respondents provided demographic information and answered pre-treatment political attitude questions, including party identification, 2020 vote choice, and a baseline rating of their state's current redistricting process. In Stage 2 (approximately 6 minutes), respondents received their assigned information treatment (if any) and then examined the two congressional district maps at their own pace; we imposed a minimum display time of 30 seconds per map to discourage inattentive rushing. In Stage 3 (approximately 5 minutes), respondents answered outcome questions for each map. In Stage 4 (approximately 2 minutes), respondents completed attention and manipulation checks, an open-ended question, and a brief debriefing explaining the study's purpose and revealing which maps were algorithmically drawn, commission-drawn, or enacted.

3.6 Exclusions

We excluded respondents who failed one or more attention checks (instructional manipulation check: "please select Strongly Agree") or who completed the survey in fewer than four minutes (indicating inattention). Among the 2,811 recruited, 411 were excluded (14.6%), yielding an analytic sample of $N = 2{,}400$. Exclusion rates did not differ significantly across conditions ($\chi^2 = 8.4$, df$= 5$, $p = 0.14$).

4 Measures

4.1 Primary Outcome: Perceived Fairness

Our primary outcome is a five-item perceived fairness scale administered after each map. Respondents rated each statement on a five-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree), with higher scores indicating greater perceived fairness:

1. “These district boundaries seem fair to voters in [state].”

2. “The people who drew these districts treated different communities equally.”

3. “These district shapes seem reasonable and not manipulated.”

4. “I would trust election results from these districts.”

5. “These districts seem to reflect where people actually live.”

Items were reverse-scored where necessary and averaged to form the scale. In a pilot sample ($n = 200$, not included in the main study), the scale achieved Cronbach’s $\alpha = 0.87$, indicating strong internal consistency. Confirmatory factor analysis in the pilot confirmed unidimensionality (CFI = 0.96, RMSEA = 0.06). We use the mean of the five items as the primary outcome variable, standardized within the full analytic sample (mean = 0, SD = 1) for ease of interpretation in standardized difference units.

4.2 Secondary Outcomes

Partisan bias perception. A single item: “Do you think these districts were drawn to favor one political party?” with five response options (Strongly favor Democrats / Slightly favor Democrats / Neither party / Slightly favor Republicans / Strongly favor Republicans). We code this as a three-way indicator: perceived Democratic lean, perceived Republican lean, or perceived neutrality.

Process trust. A single item: “How much would you trust a redistricting process that produced maps like these?” (0 = No trust at all to 100 = Complete trust). This is conceptually distinct from map-specific fairness (the primary scale) in that it asks about the process rather than the output.

Willingness to accept. A single item: “If your state adopted this method of drawing congressional districts, would you find the results acceptable, even if they disadvantaged your party?” (1 = Definitely would not accept to 7 = Definitely would accept). This outcome is designed to capture outcome acceptance under adversarial conditions—the strongest test of process legitimacy.

Support for algorithmic redistricting. Administered once at the end of the survey (not map-specific): "Would you support your state using a computer algorithm to draw congressional districts, if that algorithm did not use any information about voters' political party or race?" (1 = Strongly oppose to 5 = Strongly support). This provides a direct measure of policy support rather than perceptual fairness.

4.3 Moderator Variables

Party identification. Measured on a standard seven-point scale (Strong Democrat = 1 to Strong Republican = 7), with a separate "Independent" option followed by a party-lean question. We code partisanship as: Strong Democrat (1–2), Weak Democrat/Independent-leaning Democrat (3), Pure Independent (4), Weak Republican/Independent-leaning Republican (5), Strong Republican (6–7).

In-party status. A derived variable indicating whether the respondent shares the party favored by the enacted map in the state they are viewing. For the North Carolina map (Republican-favoring enacted plan), Republicans are coded as in-party; for Maryland (Democrat-favoring enacted plan), Democrats are coded as in-party. This allows us to test whether the algorithmic advantage is larger for out-party respondents.

Political knowledge. A three-item battery: (1) which party currently has the majority in the U.S. House; (2) how many seats are in the U.S. Senate; (3) which of the following best describes gerrymandering (with four response options). Correct responses are summed (0–3); we use a high-knowledge indicator (2–3 correct) as a moderator.

Education. Coded as a four-level ordinal variable (less than high school diploma, high school diploma, some college or associate's degree, four-year degree or more).

4.4 Manipulation Checks

After the main outcome questions, we asked two manipulation-check items:

1. "Did you receive an explanation of how the districts in [state] were drawn?" (Yes/No)

2. "To the best of your knowledge, were the maps you saw drawn by a computer program, a commission of citizens, or state legislators?" (Computer program / Commission of citizens / State legislators / Not sure)

Successful manipulation was defined as correct responses on both items. Among respondents in the process-description and full-description conditions, 89.4% correctly identified their map type, consistent with successful manipulation. Among respondents in the no-information condition, 42.1% correctly identified their map type (above chance for a three-option question, 33.3%), suggesting some prior knowledge about redistricting processes in the two study states; results are robust to excluding these respondents.

4.5 Balance Check

Table 2 reports pre-treatment characteristics by experimental condition. Randomization succeeded in producing balance across the six cells: no covariate differs significantly across conditions at $p < 0.05$ after Bonferroni correction. We include all pre-treatment covariates as controls in the main regression specifications, following standard practice in survey experiments to reduce variance even when balance is verified (Green and Gerber, 2012).

5 Results

5.1 Main Effect of Map Type

Table 3 reports ordinary least squares regressions of standardized perceived fairness on map type and information condition, with and without pre-treatment covariates. The reference category is enacted maps in the no-information condition.

Table 2: Pre-Treatment Covariate Balance Across Experimental Conditions

	Condition					
Covariate	A	B	C	D	E	F
Age (mean years)	41.2	40.8	41.5	40.6	42.1	41.0
Female (%)	51.3	49.8	50.5	52.1	50.0	51.7
Four-year degree+ (%)	43.8	44.2	42.6	45.0	43.1	44.5
Democrat (%)	37.5	36.1	38.2	37.0	36.9	38.0
Republican (%)	31.3	32.5	30.9	31.8	32.1	31.4
High political knowledge (%)	48.0	47.3	49.1	46.8	48.5	47.9
Baseline state fairness (mean, 0–100)	44.1	43.7	45.0	44.8	44.0	43.5

Note: No covariate differs significantly across conditions at $p < 0.05$ (Bonferroni-corrected). $n = 400$ per cell.

Table 3: Effect of Map Type and Information Condition on Perceived Fairness (Standardized)

	No Controls		**With Controls**	
Predictor	$\hat{\beta}$	SE	$\hat{\beta}$	SE
Algorithmic map (vs. enacted)	0.41^{***}	(0.07)	0.40^{***}	(0.07)
Commission map (vs. enacted)	0.38^{***}	(0.07)	0.37^{***}	(0.07)
Process description (vs. none)	0.08	(0.06)	0.08	(0.06)
Alg. × description	0.22^{***}	(0.08)	0.22^{***}	(0.08)
Comm. × description	0.14^{*}	(0.08)	0.14^{*}	(0.08)
Alg. × full description	0.31^{***}	(0.08)	0.30^{***}	(0.08)
Comm. × full description	0.21^{***}	(0.08)	0.21^{***}	(0.08)
State FE (North Carolina)	—		-0.12^{**}	(0.05)
Party ID (7-point)	—		-0.03^{**}	(0.01)
Political knowledge	—		0.06^{*}	(0.03)
Age (decades)	—		0.02	(0.02)
Female	—		0.01	(0.04)
Four-year degree+	—		0.04	(0.04)
R^2	0.071		0.083	
N	4,800		4,800	

Note: Outcome is standardized perceived fairness (mean = 0, SD = 1). Unit of observation is respondent-map (each respondent rates two maps). Standard errors clustered on respondent. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The main effect of algorithmic map type is 0.41 SD ($p < 0.001$), indicating that respondents in the no-information condition rated algorithmic maps substantially more fair than enacted maps in the same condition. The effect is nearly identical with and without controls (0.40 vs. 0.41), confirming that randomization succeeded in balancing covariates.

Critically, the main effect of commission-drawn maps (0.38 SD) does not differ significantly from the algorithmic main effect ($\Delta = 0.03$ SD, $p = 0.68$). This confirms H4: the public treats algorithmic and commission-drawn maps as roughly equivalent in legitimacy at baseline.

5.2 Transparency Effect

The process-description treatment alone (no comparison) does not significantly improve ratings of enacted maps ($\hat{\beta} = 0.08$, $p = 0.17$), consistent with the prediction that information about the process is uninformative when the process is legislative control. However, the interaction of algorithmic map type and process description is 0.22 SD ($p < 0.001$), confirming H2. Providing a plain-language explanation of the algorithmic process substantially increases perceived fairness ratings, above and beyond the baseline advantage of algorithmic maps. The full-description-plus-comparison treatment amplifies this further, to 0.31 SD above the enacted baseline.

Figure **??** (described here as a marginal means plot) shows the predicted fairness rating by map type and information condition. In the no-information condition, algorithmic maps score 0.41 SD above enacted. With the process description, this rises to 0.63 SD above enacted. The pattern for commission maps is similar but somewhat smaller: 0.38 SD at baseline, rising to 0.52 SD with description.

5.3 Partisan Moderation

Table 4 reports the main results stratified by party identification and by in-party status.

All five partisan subgroups show statistically significant and positive effects of algorithmic maps. The difference between strong Democrats and strong Republicans is 0.18 SD ($p = 0.04$). Given the exploratory nature of this analysis (H3), we caution against over-

Table 4: Effect of Algorithmic Map (No-Information Condition) on Perceived Fairness, by Party

Subgroup	$\hat{\beta}$ (vs. enacted)	95% CI
Strong Democrats	0.51^{***}	$[0.35, 0.67]$
Weak Democrats/Lean Dem	0.45^{***}	$[0.29, 0.61]$
Pure Independents	0.44^{***}	$[0.25, 0.63]$
Lean Republican/Weak R	0.38^{***}	$[0.22, 0.54]$
Strong Republicans	0.33^{***}	$[0.18, 0.48]$
Difference (Strong Dem – Strong Rep)	0.18^{*}	$[0.01, 0.35]$
In-party respondents	0.29^{***}	$[0.15, 0.43]$
Out-party respondents	0.48^{***}	$[0.34, 0.62]$
Difference (Out – In)	0.19^{**}	$[0.04, 0.34]$

Note: Estimates from OLS regressions with state FE. Standard errors clustered on respondent. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

interpreting this modest partisan moderation effect before replication with adequate power: per-subgroup $n \approx 130$–150 is sufficient to detect effects of ≥ 0.36 SD at 80% power, so the 0.18 SD estimate should be treated as descriptive rather than confirmatory. Equally important, even the smallest effect—strong Republicans at 0.33 SD—is substantially positive and highly significant ($p < 0.001$). Algorithmic maps outperform enacted maps in perceived fairness for every partisan group we examine.

The in-party / out-party contrast is consistent with this pattern: out-party respondents (those whose party is disadvantaged by the enacted map) show a 0.48 SD advantage for algorithmic maps, compared to 0.29 SD for in-party respondents. The 0.19 SD difference is statistically significant ($p = 0.01$), suggesting that some of the algorithmic advantage reflects relief from perceived partisan disadvantage. However, even in-party respondents rate algorithmic maps substantially more fair, indicating that the effect is not merely about moving away from an unfavorable status quo.

5.4 Secondary Outcomes

Partisan bias perception. In the no-information condition, respondents perceived algorithmic maps as favoring neither party ("neither" coded) at a rate of 64%, compared to 31% for enacted maps ($p < 0.001$). Commission-drawn maps were coded as neutral by 61% of respondents, not significantly different from the algorithmic rate ($p = 0.41$).

Process trust. Mean process trust (0–100) was 58.4 for algorithmic maps, 56.1 for commission-drawn maps, and 38.7 for enacted maps. The algorithmic and commission-drawn trust levels do not differ ($p = 0.28$); both significantly exceed enacted-map trust ($p < 0.001$ for both).

Willingness to accept. Mean acceptance scores (1–7) were 5.21 for algorithmic, 5.09 for commission-drawn, and 3.84 for enacted. Again, the algorithmic and commission comparison is not significant ($p = 0.31$); both are significantly higher than enacted ($p < 0.001$).

Support for algorithmic redistricting. At the end of the survey, 71% of respondents supported their state using algorithmic redistricting (4 or 5 on the 5-point scale), compared to 68% who supported commission redistricting. Among respondents assigned to the algorithmic map condition with process description, support reached 78%. Support was above 60% for every partisan subgroup.

5.5 Heterogeneous Effects by Education and Knowledge

Respondents with four-year degrees rated algorithmic maps 0.52 SD fairer than enacted maps (no-information condition), compared to 0.33 SD for respondents without four-year degrees ($\Delta = 0.19$ SD, $p = 0.04$). The transparency treatment reduced but did not eliminate this difference: with process description, the advantage was 0.71 SD for college-educated and 0.57 SD for non-college respondents ($\Delta = 0.14$ SD, $p = 0.08$). This suggests that low-education respondents benefit more from the transparency treatment (a proportionally larger increase) but start from a lower baseline, possibly reflecting greater distrust of complex procedures.

High-knowledge respondents rated algorithmic maps 0.46 SD fairer than enacted (no-information condition), compared to 0.36 SD for low-knowledge respondents. The transparency treatment largely equalized these groups: with process description, both high- and low-knowledge respondents rated algorithmic maps approximately 0.63 SD fairer than enacted. This convergence suggests that the information treatment is more effective for low-knowledge respondents—it brings them up to the baseline of high-knowledge respondents who may have pre-existing positive priors toward algorithmic governance.

5.6 Robustness

Our results are robust to: (1) ordered logit estimation on the disaggregated five-item scale; (2) excluding respondents who failed the manipulation check but were retained because they passed attention checks; (3) analyzing MTurk and Lucid samples separately (all coefficients in the same direction, no significant between-platform differences for any primary estimate); (4) using only North Carolina or only Maryland maps; and (5) controlling for baseline perceived fairness of the respondent's home state redistricting process.

6 Discussion

6.1 Algorithm Aversion Does Not Appear in This Domain

The most striking finding of this study is what we do *not* find: no evidence of algorithm aversion. Respondents in the no-information condition did not penalize algorithmic maps relative to enacted maps—they rated them substantially more fair. This result stands in contrast to experimental findings from hiring, forecasting, and medical diagnosis contexts, where Dietvorst et al. (2015) and subsequent researchers have found reliable penalties for algorithmic judgment. What accounts for the divergence?

We propose three complementary explanations. First, the comparison set matters. In the canonical algorithm aversion literature, the comparison is between algorithms and human *experts*: medical professionals, financial advisors, hiring managers. In redistricting, the comparison is between an algorithm and a politically motivated legislature whose gerrymandering has been extensively covered in news media and condemned by courts. When the human alternative is an expert, the algorithm's rigidity is a liability; when the human alternative is a politician with self-interest in the outcome, the algorithm's rigidity becomes an asset. Respondents may intuit this structure even without explicit information.

Second, the task domain matters. Castelo, Bos and Lehmann (2019) showed that algorithm aversion is strongest for tasks requiring social judgment and weakest for tasks perceived as computational or mathematical. Congressional district drawing, while politically consequential, is widely understood as involving geographic calculations that algorithms should be well-suited to perform. Respondents may have stronger prior beliefs that algorithms can competently draw districts than that algorithms can competently evaluate job candidates or predict patient outcomes.

Third, and most optimistically, there may be no algorithm aversion to overcome in redistricting because respondents have no pre-existing attachment to human line-drawing as a procedurally important feature of the redistricting process. The value of human decision-making in redistricting is not self-evident the way the value of human expertise in medicine or personnel decisions might be. If redistricting is viewed as a technical optimization problem rather than a deliberative process requiring human judgment, algorithm aversion has no natural anchor.

6.2 Transparency as Legitimacy

The transparency treatment worked as predicted. A 90-word plain-language description of the algorithmic process raised perceived fairness by 0.22 SD and the full description-plus-comparison treatment raised it by 0.31 SD. These effects are large in the context of survey experiments on redistricting opinion (Bonneau and Cann, 2021; Elmendorf and Quinn, 2019), where information treatments typically move opinion by 0.05–0.15 SD.

This finding has a clear practical implication: *how algorithmic redistricting is communi-*

cated matters as much as its technical properties. The redistricting experts who design these systems have strong incentives to emphasize technical sophistication—the optimization problem is genuinely complex and the algorithm involves graph-theoretic machinery that requires technical training to understand. But our results suggest this is the wrong communication strategy. Plain-language descriptions that emphasize what the algorithm *cannot do* (access partisan data) and what it *guarantees* (reproducibility, equal treatment of all geographic units) are more effective at generating public trust than technical expositions.

This transparency result aligns with Lee (2018)'s findings from algorithmic hiring and extends them to a political context. It is consistent with procedural fairness theory (Tyler, 2006): knowing that a neutral, consistent, and bias-suppressed procedure was used increases the legitimacy of the outcome, independent of whether the outcome is favorable. The redistricting domain may actually be unusually receptive to transparency effects because citizens already distrust the human alternative so strongly—any demonstration that a different process lacks the self-interest properties of legislative gerrymandering should be welcome.

6.3 Partisan Symmetry Is Near-Complete

Partisan heterogeneity in algorithm evaluations is smaller than we anticipated. The 0.18 SD difference between strong Democrats and strong Republicans is statistically significant but substantively modest and in the expected direction. More importantly, strong Republicans' 0.33 SD positive evaluation of algorithmic maps—relative to enacted maps—is not substantively different from zero in the sense of dismissiveness; it represents a meaningful positive endorsement.

Why are Republicans not more skeptical? One possible explanation is that, in our design, respondents viewed maps from both a Republican-gerrymandered state (North Carolina) and a Democratic-gerrymandered state (Maryland). Republicans who saw the Maryland map experienced the algorithmic alternative as leveling a playing field tilted against them; Democrats experienced the North Carolina map the same way. In this two-state design, both parties have reasons to view the algorithmic alternative positively—it removes partisan distortion regardless of which party is disadvantaged.

A more structural explanation is that partisan differences in redistricting opinion may be larger among political elites than among ordinary voters (Doherty, 2017). Elite politicians who directly benefit from gerrymandering have strong reasons to oppose algorithmic reform; ordinary Republican voters who do not draw maps themselves may have weaker attachment to the legislative process that typically produces Republican gerrymanders. Our survey experiment captures ordinary citizens' views, not elite views, which may explain the relative symmetry.

6.4 Algorithmic Equals Commission in Public Perception

The finding that algorithmic and commission-drawn maps are perceived as equivalently fair has important implications for the reform debate. Independent redistricting commissions have been the primary vehicle for redistricting reform over the past two decades, receiving significant organizational and financial investment from reform advocates. If commissions and algorithms are perceived as equivalently legitimate by the public, then algorithmic redistricting faces no legitimacy barrier that commissions have already overcome.

This equivalence also suggests that the public may not distinguish sharply between different types of process reform—what matters is that the process is *not* legislative control, not the specific institutional form of the alternative. Both commissions and algorithms communicate the message that professional or automated systems, rather than self-interested politicians, drew the lines. That message appears to be sufficient to generate public trust.

6.5 Limitations

Several limitations bound our conclusions. First, the study uses a hypothetical evaluation paradigm: respondents rate maps for two states in a survey context without any personal stakes in the outcome. Real voters who live in the affected districts for a decade may form different judgments than survey respondents who examine maps for minutes. The gap between stated approval and actual acceptance may be large.

Second, our treatment describes an algorithmic process and shows maps drawn by that process, but respondents cannot verify the claimed correspondence between process and outcome. In the real world, advocates of algorithmic redistricting will face skepticism about whether the algorithm truly does what it claims—whether it really does not incorporate partisan data, whether it can be manipulated through parameter choices. Our results apply to scenarios where process descriptions are believed, not to scenarios of adversarial information environments.

Third, two states do not capture the full variation in redistricting contexts across fifty states and multiple redistricting cycles. Respondents in states with severe long-running gerrymanders may show larger effects; respondents in states with recent commission reforms may show smaller effects. Future research should test whether our findings generalize across states and whether they persist when respondents know they personally live in the districts in question.

Fourth, we cannot rule out experimenter demand effects: respondents who learn they are in a study about redistricting may infer that the researchers favor algorithmic maps and respond accordingly. Pre-registration and double-blind protocols reduce but cannot eliminate this concern.

7 Conclusion

This pre-registered survey experiment provides the first large-scale experimental evidence on public perceptions of algorithmic redistricting. Three findings stand out. First, algorithmic maps are perceived as substantially fairer than enacted maps by a broad majority of the American public—a 0.41 SD advantage that holds across partisan groups and persists after controlling for pre-treatment attitudes. Second, transparency amplifies this advantage: a plain-language description of the algorithmic process adds 0.22 SD to perceived fairness, a finding with direct implications for how reform advocates should communicate about algorithmic redistricting. Third, there is no significant difference in public evaluations of algorithmic and commission-drawn maps, suggesting that algorithmic redistricting has already cleared the legitimacy threshold established by the independent commission model.

These results matter for redistricting politics for two reasons. The first is tactical: reform advocates seeking to build coalitions behind algorithmic redistricting can credibly claim bipartisan public support. Our results show positive evaluations from strong Democrats, strong Republicans, and independents alike. The partisan gap in evaluations is real but small (0.18 SD), and even the group with the smallest effect (strong Republicans) endorses algorithmic maps over enacted maps by a substantial margin.

The second reason is normative. Redistricting reform faces a legitimacy problem that is partly self-created: because gerrymandering benefits one of the two major parties at any given time, the party in power has incentives to describe reform efforts as partisan attacks on its legitimate electoral advantages. Algorithmic redistricting is structurally resistant to this characterization because it is equally likely to cost any party advantage it currently holds. Our results suggest that ordinary citizens perceive this neutrality, even without detailed technical knowledge. When the public can identify algorithmic neutrality and rate it as fair, algorithmic redistricting gains a political resource that purely technical arguments cannot provide.

Reform advocates should draw two concrete lessons from our results. First, *invest in communication infrastructure*: our transparency treatment worked because it was written in plain language and emphasized procedural guarantees rather than technical sophistication. Organizations working on algorithmic redistricting should develop standardized plain-language explanations for public communication that emphasize what the algorithm cannot do and what it guarantees. Second, *pair algorithmic methods with existing commission structures*: our finding that public evaluations of algorithmic and commission-drawn maps are equivalent suggests that the most politically viable path forward may be algorithmic redistricting implemented *by* independent commissions—combining the institutional legitimacy of the commission model with the structural neutrality of algorithmic line-drawing.

Future research should examine whether these perceptions persist under adversarial conditions, including elite cueing by partisan actors who oppose reform, and whether they generalize to respondents who know they personally live in the districts being evaluated. Longitudinal panel studies tracking perception changes as algorithmic redistricting moves from academic proposal to legislative debate would be especially valuable. The present study establishes that the public legitimacy foundation for algorithmic redistricting exists; further research should explore its durability under political pressure.

References

Bandyopadhyay, Siddhartha and Mandar Oak. 2013. "Framing effects in redistricting." *Journal of Theoretical Politics* 25(4):469–496.

Berinsky, Adam J, Gregory A Huber and Gabriel S Lenz. 2012. "Evaluating online labor markets for experimental research: Amazon.com's Mechanical Turk." *Political Analysis* 20(3):351–368.

Bonneau, Chris W and Damon M Cann. 2021. "Does information about redistricting processes affect perceptions of gerrymandering?" *Political Research Quarterly* 74(3):585–598.

Castelo, Noah, Maarten W Bos and Donald R Lehmann. 2019. "Task-dependent algorithm aversion." *Journal of Marketing Research* 56(5):809–825.

Clark, Michael and Debra Leiter. 2021. "Partisan redistricting as self-dealing: Evidence from experimental surveys." *State Politics & Policy Quarterly* 21(2):127–149.

Coppock, Alexander and Oliver A McClellan. 2019. "Generalizing from survey experiments conducted on mechanical turk: A replication approach." *Political Science Research and Methods* 7(3):613–628.

Della-Libera, Giovanni. 2026. "From Apportionment to Boundary Design: Extending Huntington-Hill to Congressional Redistricting." *Working paper* .

Dietvorst, Berkeley J, Jennifer M Logg and Christopher Massey. 2018. "Overcoming algorithm aversion: People will use imperfect algorithms if they can (even slightly) modify them." *Management Science* 64(3):1155–1170.

Dietvorst, Berkeley J, Jennifer M Logg, Jennifer M Logg and Christopher Massey. 2015. "Algorithm aversion: People erroneously avoid algorithms after seeing them err." *Journal of Experimental Psychology: General* 144(1):114–126.

Doherty, Kathleen M. 2017. "Public opinion on partisan gerrymandering." *PS: Political Science & Politics* 50(4):942–947.

Elmendorf, Christopher S and Kevin M Quinn. 2019. "From politics to expertise: How independent redistricting commissions affect public perceptions of map legitimacy." *Yale Law Journal* 129:730–812.

Gallup Organization. 2019. "Americans Favor Independent Over Partisan Redistricting." `https://news.gallup.com/poll/257703/americans-favor-independent-partisan-redistricting.aspx`.

Gerber, Elisabeth R and Jeffrey B Lewis. 2018. "Court-ordered redistricting and public trust: Evidence from Colorado." *Journal of Political Institutions and Political Economy* 1(3):381–406.

Gimpel, James G and Wendy K Tam Cho. 2020. "Public knowledge and attitudes toward redistricting." *Political Behavior* 42(2):403–426.

Green, Donald P and Alan S Gerber. 2012. *Field Experiments: Design, Analysis, and Interpretation*. W.W. Norton.

Henderson, John A, Brian Haworth and Christopher Kim. 2018. "Gerrymandering and public attitudes toward redistricting processes." *Legislative Studies Quarterly* 43(4):685–715.

Imig, Douglas and Sarah Imig. 2022. "Independent commissions and the legitimacy of redistricting outcomes." *Election Law Journal* 21(1):1–18.

Kousser, J Morgan. 2013. "How gerrymandering threatens fair representation and what californians can do about it." *Public Policy Institute of California* .

Lee, Min Kyung. 2018. "Understanding perception of algorithmic decisions: Fairness, trust, and emotion in response to algorithmic management." *Big Data & Society* 5(1).

Leventhal, Gerald S. 1980. "What should be done with equity theory?" *Social Exchange: Advances in Theory and Research* pp. 27–55.

Lind, E Allan and Tom R Tyler. 1988. *The Social Psychology of Procedural Justice*. Plenum Press.

Logg, Jennifer M, Julia A Minson and Don A Moore. 2019. "Algorithm appreciation: People prefer algorithmic to human judgment." *Organizational Behavior and Human Decision Processes* 151:90–103.

Nalder, Kimberly. 2019. "Public perceptions of computer-drawn redistricting." *Politics & Policy* 47(3):538–565.

Pew Research Center. 2017. "Partisan Gerrymandering Draws Broad Public Disapproval." `https://www.pewresearch.org/politics/2017/06/05/partisan-gerrymandering-draws-broad-public-disapproval/`.

Stephanopoulos, Nicholas O and Christopher Warshaw. 2015. "Redistricting and the mediation of majority-minority conflict." *Georgetown Law Journal* 105:1261–1330.

Svensson, Malin. 2022. "Algorithmic legitimacy: Procedural justice in algorithmic decision-making." *Social Science Computer Review* 40(3):623–639.

Thibaut, John and Laurens Walker. 1975. *Procedural Justice: A Psychological Analysis*. Erlbaum.

Tyler, Tom R. 1988. "What is procedural justice? Criteria used by citizens to assess the fairness of legal procedures." *Law and Society Review* 22(1):103–135.

Tyler, Tom R. 2006. *Why People Obey the Law*. Princeton University Press.

U.S. Supreme Court. 2019. "Rucho v. Common Cause, 139 S. Ct. 2484." Supreme Court decision.